INTERNSHIP REVIEW

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COLLEGE: VELLORE INSTITUTE OF TECHNOLOGY BHOPAL

COMPANY: NOKIA

ABOUT NOKIA

A B2B technology innovation leader, realizing the potential of digital in every industry.

Purpose
"At Nokia, we create technology that helps the world act together."

Driven by this purpose, we pioneer the future where networks meet cloud to unlock the full potential of digital in every industry.

Brand Promises future-ready performance , impact at scale, collaborative advantage

INTERNSHIP OVERVIEW

Role: Student Intern

Duration: 18th August ,2025 – 30th June ,2026

Department: MI CNS P&E IPT Product System Testing

Team: CFX PST System Testing

Manager: Deepak SG (IMS & NextGen Voice Product Test Leader)

Guide: Harish Sashikumar

TECHNICAL SKILLS GAINED

Category	Skills Gained
Telecom	VoLTE/IMS, Diameter/DRA, SIP, cellular evolution
IMS	Architecture , protocols, call-flows, CSCF, OAM
Product	Routing, Lawful Interception, installation & upgrade
Kubernetes	Architecture, Helm, storage, networking, monitoring, security
Automation	Robot Framework, CI/CD, Python (pandas, paramiko, subprocess)
Emerging	Machine Learning basics and AI

APPLICATION-SIDE WORK

Product Deployment in Test Environment

We create configuration files for CFX-5000 and use Podman and Docker to onboard images and charts to the Quay/Harbor registry. Deployment is triggered using Nokia's LCM tool, and we perform health checks after successful deployment.

1. **First deployment (Dec 17, 2025)**

What: Triggered deployment of the product in the testbed using the LCM tool.

Outcome: Deployment completed successfully.

Learning: Gained experience in deploying the product on a CNF (Cloud-Native Network Function) environment.

APPLICATION-SIDE WORK

2. Second Deployment (Dec 31, 2025)

What: Triggered deployment of CFX-5000 using the automation tool.

Outcome: Deployment completed successfully on CNF.

Learning: Resolved registry name mismatch; updated configuration, redeployed, and scaled up resources.

3. Onboard resource analysis (Jan 2, 2026)

What: Analyzed how to onboard images and charts to the registry for deployment.

Outcome: Was able to onboard all the images and charts to harbor registry

Learning: Understood how to upload images and charts to harbor registry.

APPLICATION-SIDE WORK

4. Kubernetes-Based Deployment (Mar 4, 2026)

What: K8s-based (Helm-based) deployment of CFX-5000 on CNF.

Tasks: Uploaded images and charts to the registry; configured CFX-5000 configuration files; deployed using Helm commands; configured Security Files; created Secrets, ConfigMaps, Helm repo, and NCM app.

Outcome: CFX-5000 successfully deployed in the test environment.

Learning: Gained experience in Helm-based deployment flow, values files, security configuration, and Kubernetes resources (Secrets, ConfigMaps).

APPLICATION-SIDE WORK

Current Work

- **UE registration and calls** – Configure and verify successful registration and calls between two UEs.
- **Troubleshooting** – Diagnose and fix failures when they occur.
- **ISSU load tests** – Bring up basic load for In-Service Software Upgrade testing.

Platform configuration (Mar 10, 2026)

What: Configured the deployment platform for future deployments.

Task: Created a configuration file with the required settings for deployment.

AUTOMATION WORK

Test Automation

Automated 6 regression test cases using Robot Framework and Python.

Libraries: paramiko, subprocess.

Benefits: Faster execution, repeatable runs, fewer manual errors, easier regression testing, and better coverage.

Python Scripts :

Dec 4, 2025: Secret and ConfigMap creation script with validation and version handling.

Dec 16, 2025: Operator-based Helm installation command in Python.

Thank You